Universidade do Minho

Ano Letivo: 2025/26

Databases

Logical Modeling

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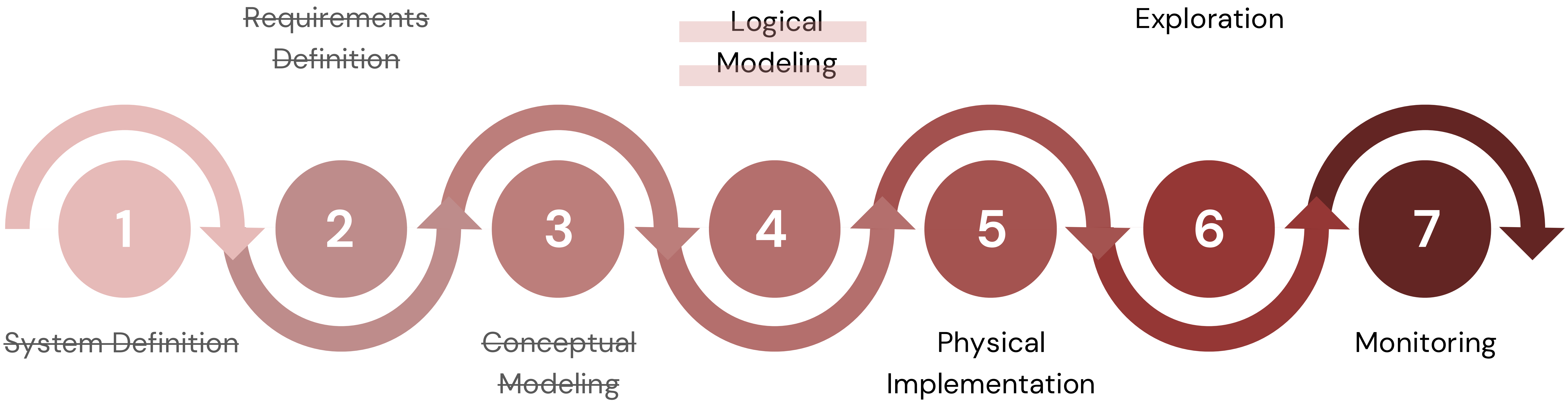
Summary

Bibliografia:

- Connolly, T., Begg, C., Database Systems, A Practical Approach to Design, Implementation, and Management , Addison-Wesley, 4a Edição, 2004. **(Chapter 17)**
- Teorey, T., Database Modeling and Design: The Fundamental Principles, II Ediçao, Morgan Kaufmann, 1994.

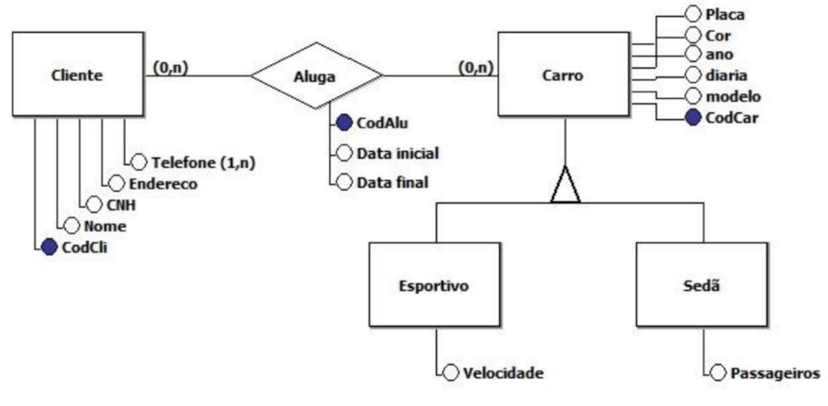
Database Development Cycle



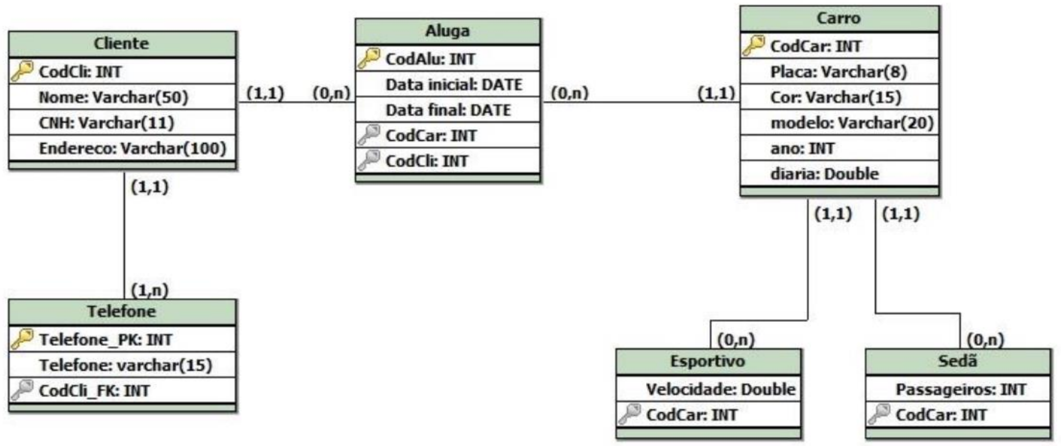
Database Development Cycle: Logical Modeling

- The logical design of a database consists of defining an information model based on a specific data model. However, this step is carried out independently of any database management system (DBMS) or aspects related to the physical implementation.

- It is based on translating the conceptual data model into a logical data model, and then validating the model to verify that it is structurally correct and able to support the required transactions.



Conceptual Model



Logic Model

Database Development Cycle: Logical Modeling

Step 1

Derive Relationships for the Logical Data Model

Step 2

Validate relationships using normalization

Step 3

Validate relationships against user transactions

Step 4

Check health constraints

Step 5

Review the logical data model with the user(s)

Step 6

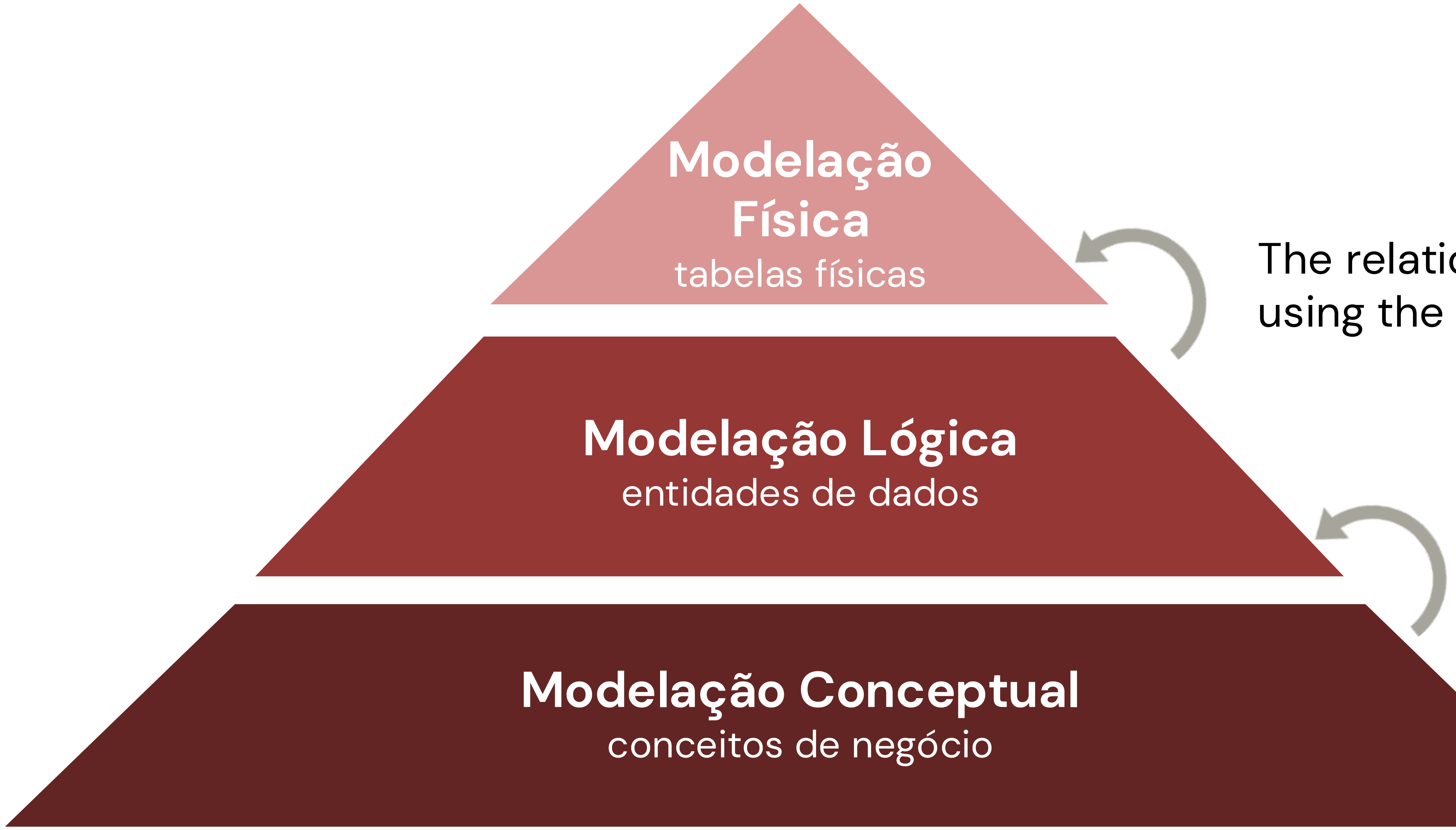
Combine logic models in a global logic model (optional)

Step 7

Check for future growth

Relational Model

Logical model for relational DBs, based on the concept of relation, also called table.



The relational model can then be realized in a DBMS using the SQL language.

The type entities and relationships of the ER model are mapped to relations/tables in the relational model.

Relational Model

- Proposed by Codd (IBM Research Fellow) in 1970 (Codd, 1970).
- A database based on the relational model of data is called a relational database.
- It is based on the concept of relation, where a relation is a **table** of values.
- A table of values can be viewed as a set of **rows or tuples.**
- Each tuple is identified by a set of **columns or attributes**.
- A database is represented as a set of relations.

relation's name

attributes

relation

ALUNO	NumMec	Nome	Curso
	798764544	João Pinto	CC
	345673451	Carlos Semedo	ERSI
	487563546	Maria Silva	EG
	452212348	Pedro Costa	MAT

tuples

PHASE 4: Logical Modeling

Conceptual-Logical Mapping

The process of derivation involves describing how relationships are derived for the following structures that can occur in a conceptual data model:

- Simple Entities
- Multivalue attributes
- Weak Entities
- One-to-many binary relationships (1:N)
- Binary Many-to-Many (N:M) Relationships
- Entity Relationship
- One-to-one (1:1) binary relationships
- One-to-one (1:1) recursive binary relationships
- Complex relationships
- Superclass/Subclass Relationships

PHASE 4: Logical Modeling

➜ Conceptual-Logical Mapping

- In converting from conceptual schema to logical schema, the first step is the simplest operation: **converting the entities into base tables.**

- All principal entities in the conceptual model now correspond to a table in the logical model. However, this correspondence is not always immediate or entirely direct.

- It is important to pay special attention to the multivalued attributes of an entity, as their mapping to the logical model requires a specific treatment.

PHASE 4: Logical Modeling

Mapeamento Conceptual-Lógico

- The relationship that an entity has with another entity is represented by the **primary/foreign key mechanism.**

- To decide where to place the foreign key attribute(s), we must first identify the 'parent' and 'child' entities involved in the relationship.

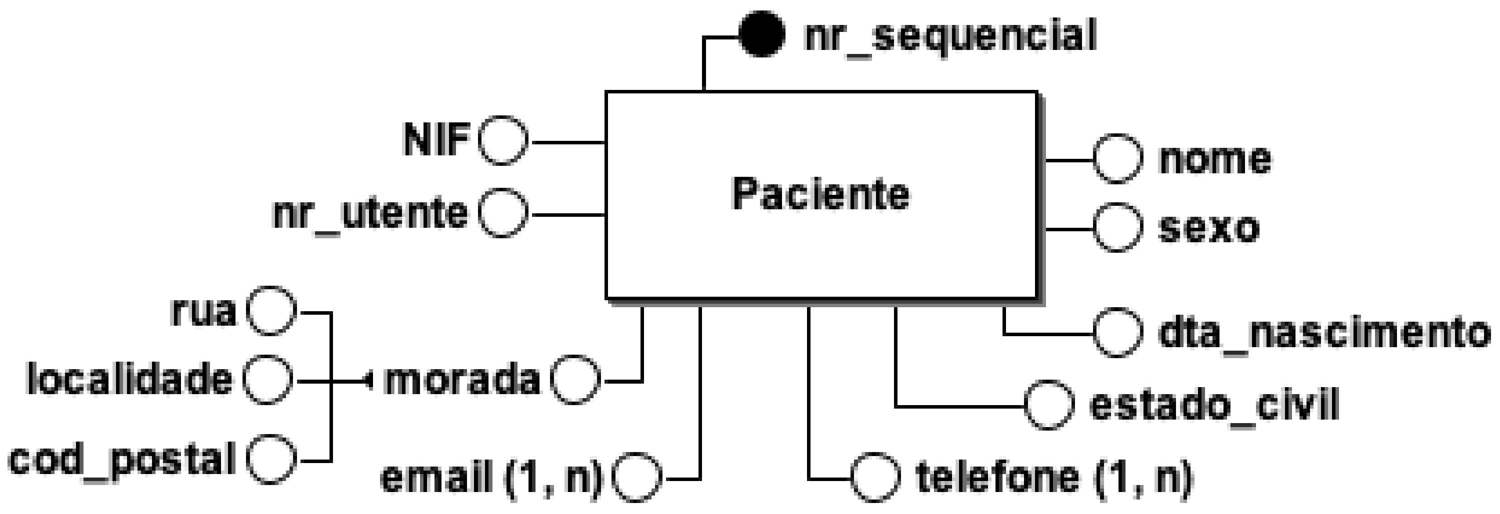
- The parent entity refers to the entity that sends a copy of its primary key in the relationship that represents the child entity, to act as the foreign key.

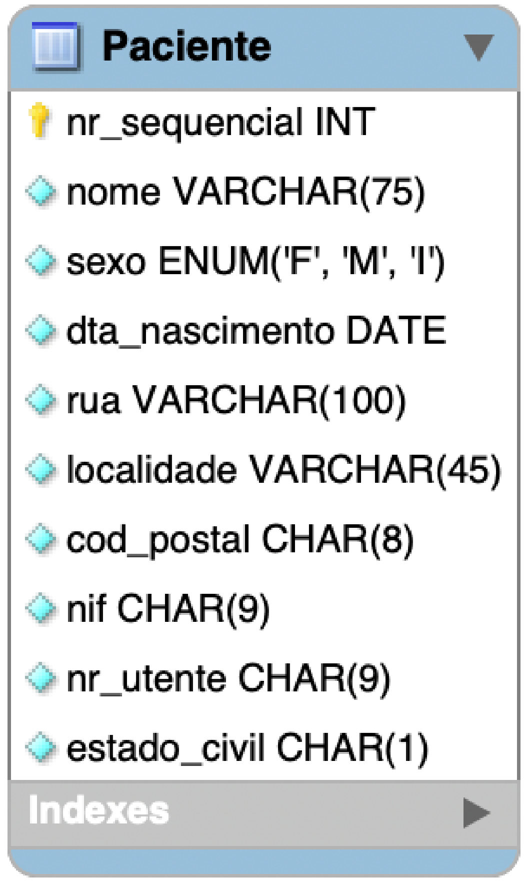
PHASE 4: Logical Modeling

Derive Relations

- **Simple Entities**

For each entity, create **a relationship/table** that includes all of the **simple** attributes of that entity. In compound attributes, only the simple constituent attributes are included.





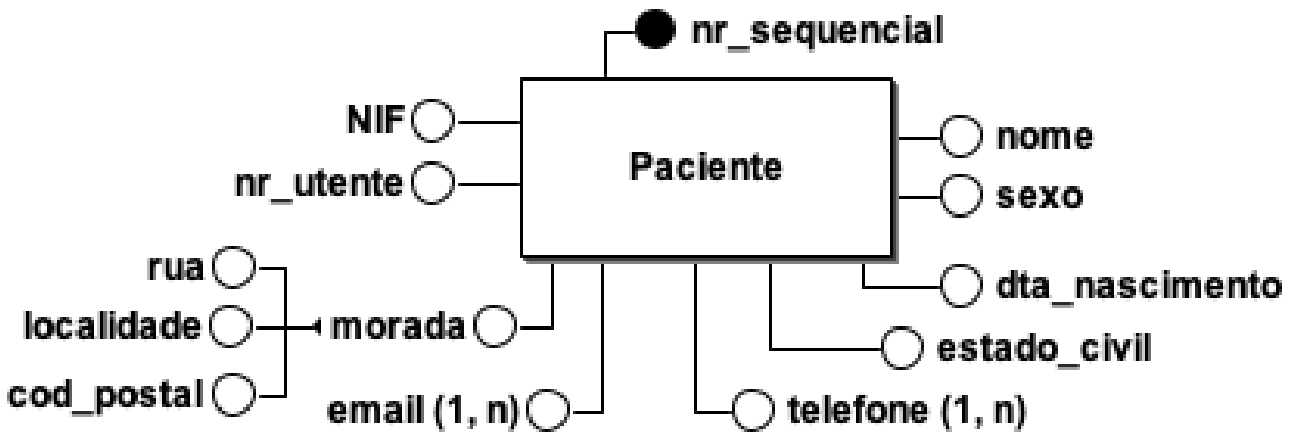
Patient (nr_sequencial, name, gender, dta_nascimento, street, city, cod_postal, NIF, nr_utente, estado_civil)
Primary key nr_sequencial
Candidate key NIF
Candidate key nr_utente

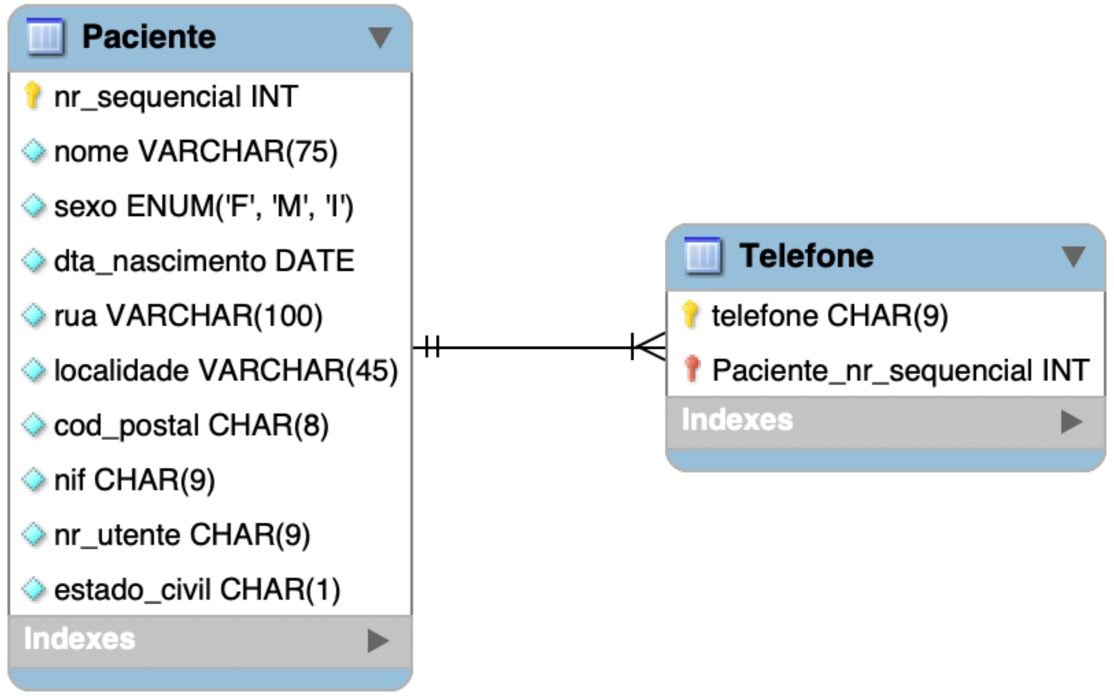
PHASE 4: Logical Modeling

➡ Derive Relations

- **Multivalue attributes**

For each **multivalued** attribute, create a **new table** with a relationship of **1:N** with your reference table, and include the entity's primary key in the new relation to act as the foreign key.





Patient (nr_sequencial, name, gender, dta_nascimento, street, city, cod_postal, NIF, nr_utente, estado_civil)
Primary key nr_sequencial
Candidate key NIF
Candidate key nr_utente

Telephone (nr_sequencial, telephone)
Primary key nr_sequencial, phone
Foreign key nr_sequencial references Patient(nr_sequencial)

PHASE 4: Logical Modeling

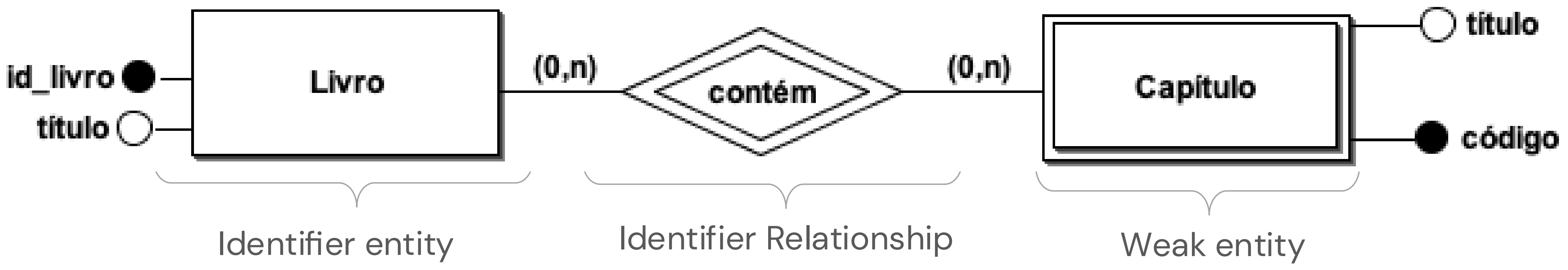
➔ Derive Relations

- **Weak Entities**

- For each weak entity in the data model, create a table that includes all of the simple attributes of that entity.

- If the weak entity does not have attributes that could constitute candidate keys, the set of attributes that allow an occurrence of the weak entity to be identified univocally is the partial key of the weak entity;

- The primary key of a weak entity is always a composite key of the primary key of the identifying entity and its partial key, so identification of the primary key of a weak entity cannot be done until all relationships with the owning entities have been mapped.



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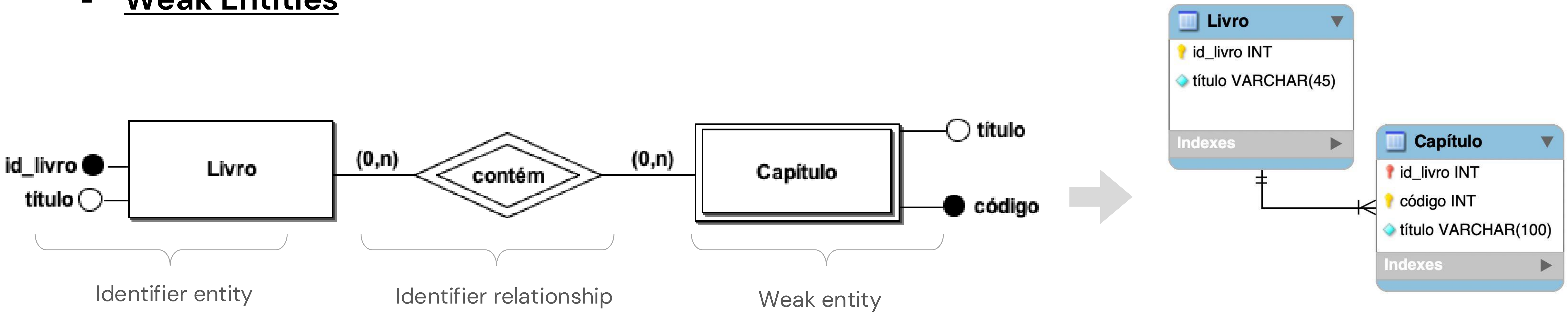
Derive Relations

- **Weak Entities**



Book (<u>id_livro</u>, title)
Primary key id_livro

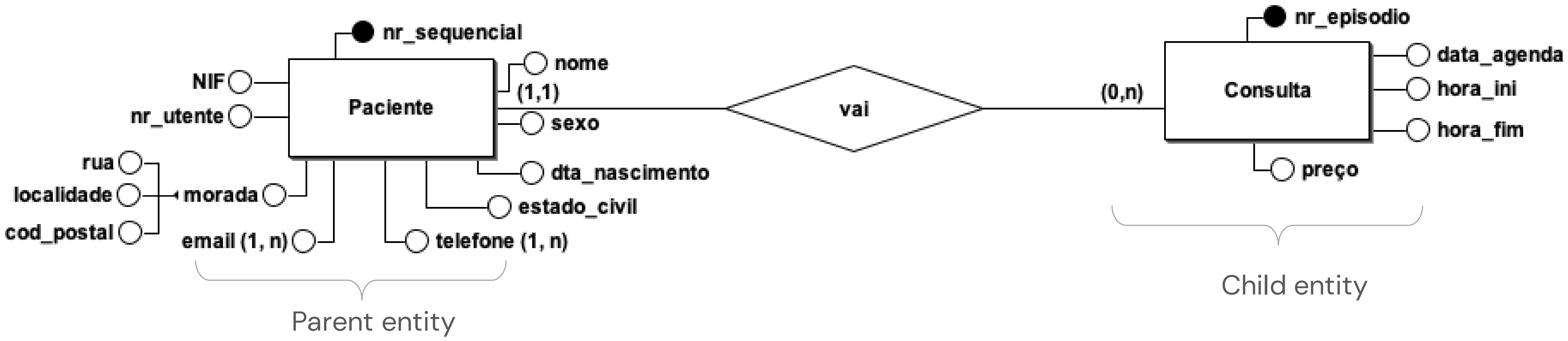
Chapter (<u>id_livro, code</u>, title)
Primary key id_livro, code
Foreign Key id_livro References Book(id_livro)

PHASE 4: Logical Modeling

Derive Relations

- **One-to-many binary relationships (1:N)**

- For each 1:N binary relationship, the entity on the '**one**' side of the relationship is designated as the **parent** entity, and the entity on the '**many**' side is designated as the **child** entity.
- To represent this relationship, a copy of the **parent** entity's **primary key** attribute(s) is created in the relations that represents the **child** entity, to act as the **foreign key**.



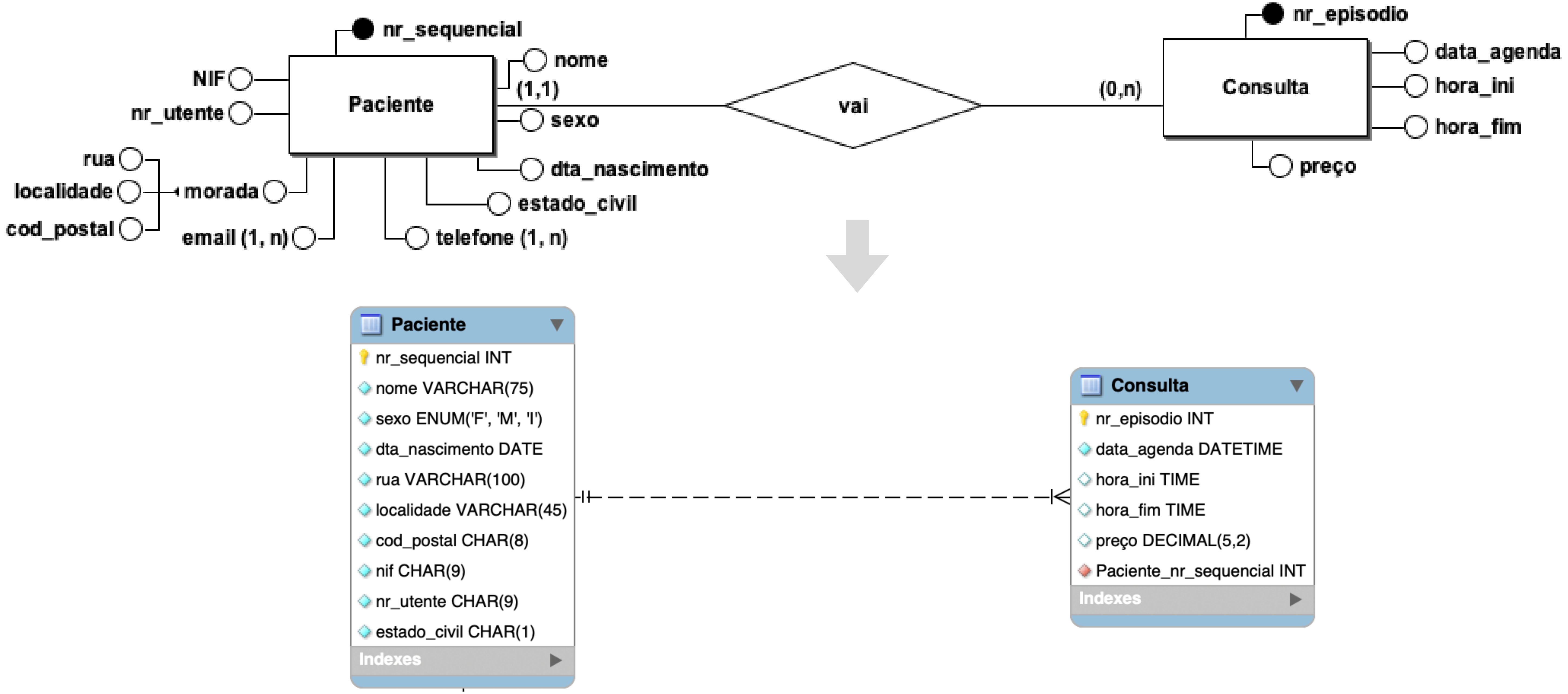
Paciente (nr_sequencial, nome, ..., estado_civil)
Primary key nr_sequencial
Candidate key NIF
Candidate key nr_utente

Consulta (nr_episodio, preço, hora_ini, hora_fim, data_agenda, id_paciente)
Primary key nr_episodio
Foreign Key id_paciente **references** Paciente(nr_sequencial)

PHASE 4: Logical Modeling

➡ Derive Relations

- One-to-many binary relationships (1:N)

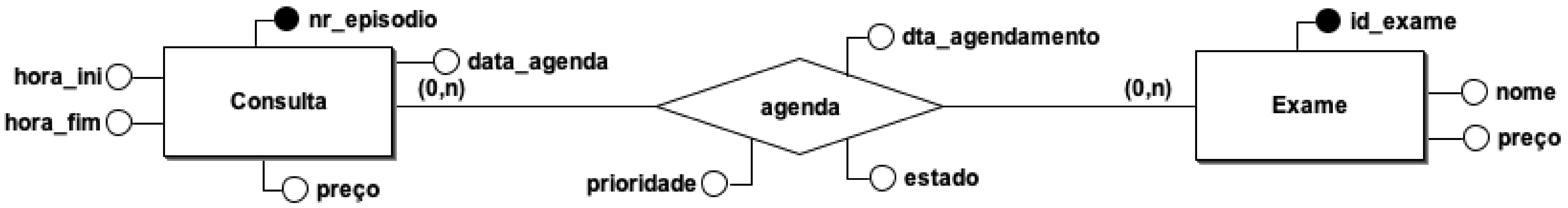


PHASE 4: Logical Modeling

Derive Relations

- **Binary Many-to-Many (N:M) Relationships**

- Create a table for the relationship and include the attributes that are part of the relationship.
- Create a copy of the primary key attribute(s) of the entities participating in the relationship in the new table, to act as foreign keys. The primary key of the new tabçe is always a key composed of the foreign keys.



Consulta (nr_episodio, data_agenda, hora_ini, hora_fim, preço)
Primary key nr_episodio

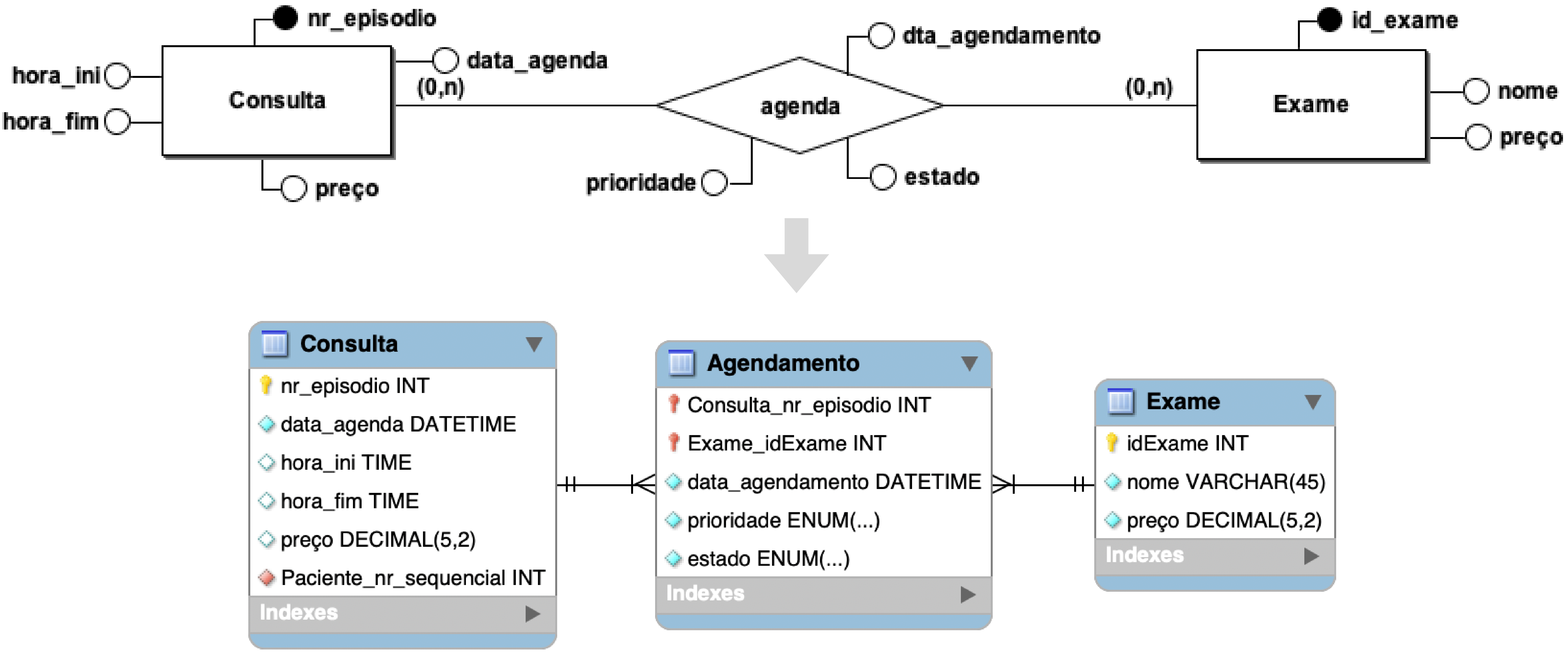
Exame (id_exame, nome, preço)
Primary key id_exame

Agendamento (id_consulta, id_exame, dta_agendamento, prioridade, estado)
Primary key id_consulta, id_exame
Foreign Key id_consulta **references** Consulta(nr_episodio)
Foreign Key id_exame **references** Exame(id_exame)

PHASE 4: Logical Modeling

Derive Relations

- **Binary Many-to-Many (N:M) Relationships**



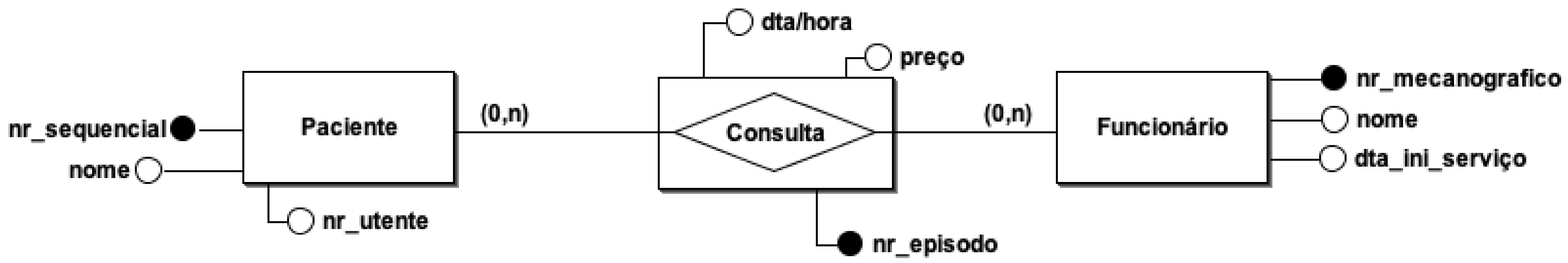
PHASE 4: Logical Modeling

Derive Relations

- **Entity Relationship**

- Create a table to represent the entity-relationship as if it were an independent entity, and include all the attributes that are part of the entity-relationship.

- Create a copy of the primary key attribute(s) of the entities that participate in the entity-relationship in the new table, to act as foreign keys. If the entity-relationship does not have a primary key, these foreign keys form the primary key.



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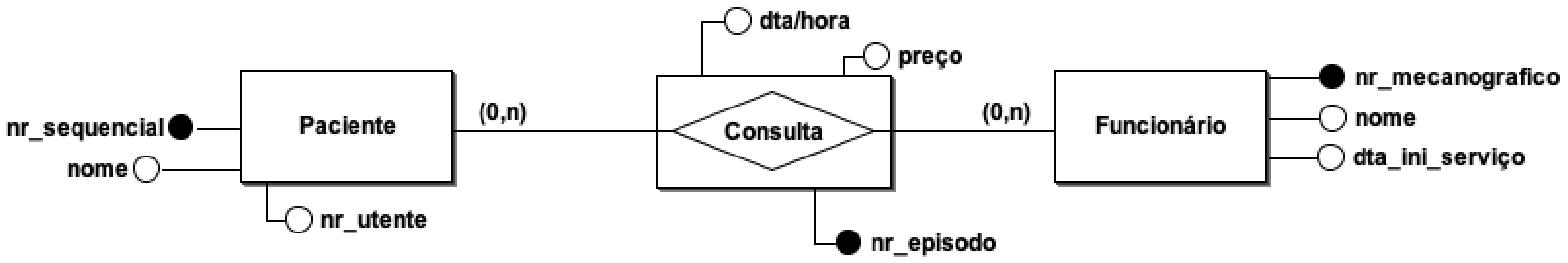
➜ Derive Relations

- **Entity Relationship**



Paciente (nr_sequencial, nome, nr_utente)
Primary key nr_sequencial
Candidate key nr_utente

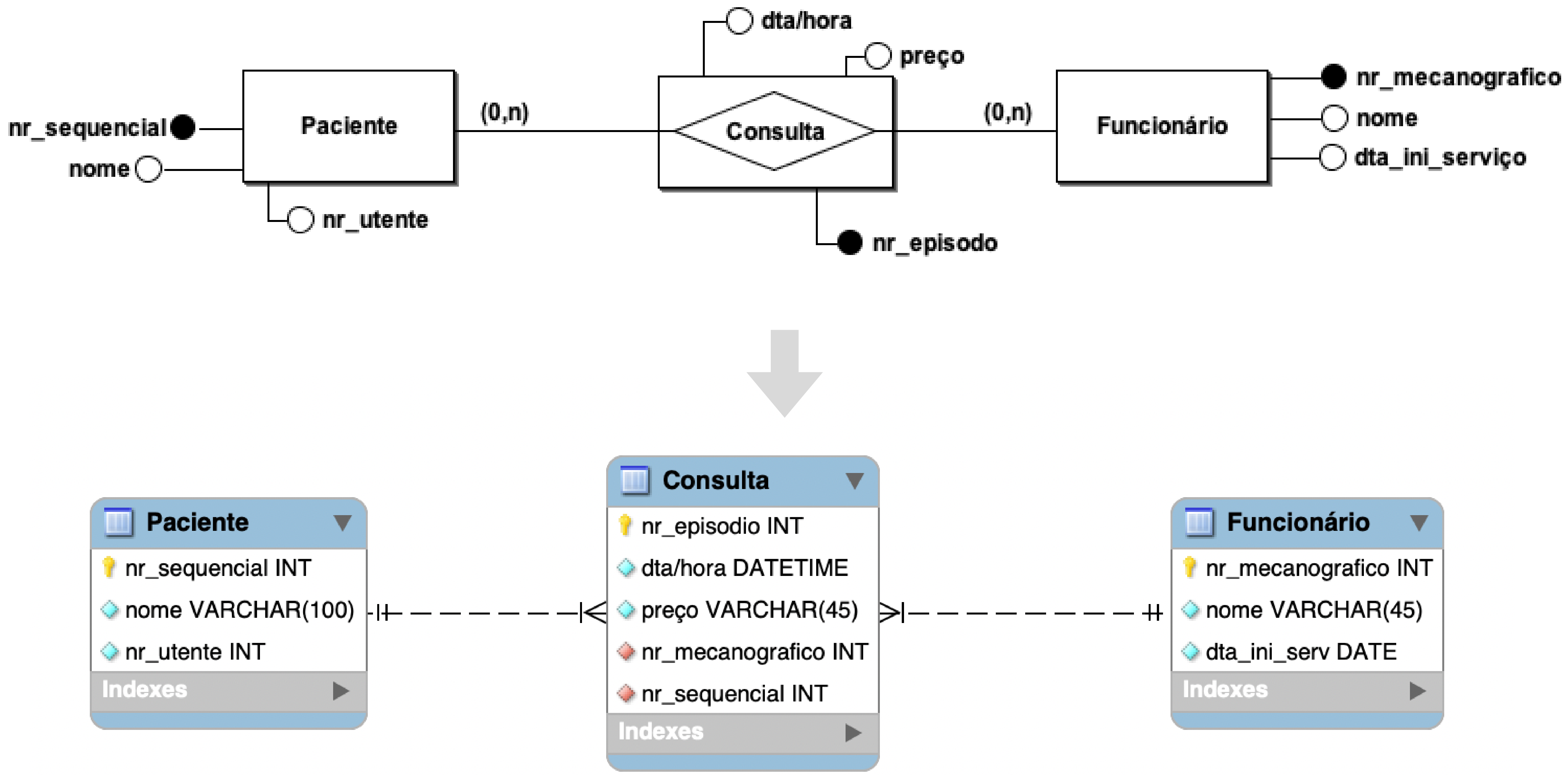
Funcionário (nr_mecanografico, nome, dta_ini_servico)
Primary key nr_mecanografico

Consulta (nr_episodio, nr_sequencial, nr_mecanografico, dta/hora, preço)
Primary key nr_episodio
Foreign Key nr_sequencial **references** Paciente(nr_sequencial)
Foreign Key nr_mecanografico **references** Funcionário(nr_mecanografico)

PHASE 4: Logical Modeling

Derive Relations

- **Entity Relationship**



PHASE 4: Logical Modeling

Derive Relations

- **One-to-one (1:1) binary relationships**

- In these cases, creating tables is more complex, because cardinality cannot be used to identify the parent and child entities in a relationship.

- Instead, participation constraints are used to decide whether it's preferable to combine entities into a single table, or whether it's more appropriate to create two tables and place a copy of the primary key from one table in the other:

  (a) mandatory participation on both sides of the 1:1 relationship;
  (b) mandatory participation on one side of the 1:1 relationship;
  (c) optional participation on both sides of the 1:1 relationship.

PHASE 4: Logical Modeling

Derive Relations

- **Re-examine one-to-one relationships (1:1)**
  - In the process of entity identification, it can happen that two entities that represent the same object in an organization's data model are detected.
  - When this occurs, these two entities must be merged into a single entity.
  - If they have different primary keys, one must be selected to assume the role of primary key, while the other is considered an alternative key.
  - Although they are uncommon and sometimes difficult to identify relationships, binary 1:1 relationships do exist.

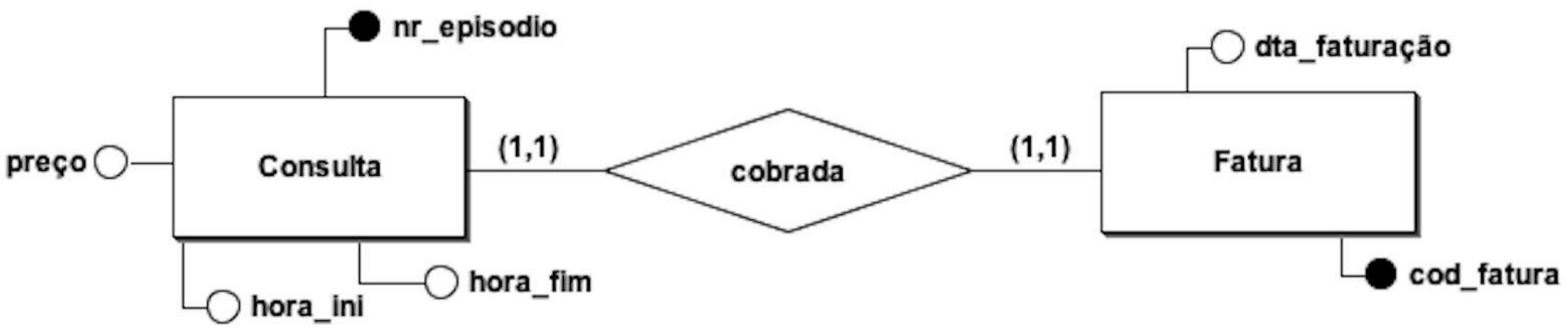
PHASE 4: Logical Modeling

Derive Relations

- **One-to-one (1:1) binary relationships**

(a) mandatory participation on both sides of the 1:1 relationship;

- Combine the entities into a single table and choose one of the primary keys to be the primary key of the new table, while the other (if it exists) is used as the candidate key.



Consulta (<u>nr_episodio</u>, preço, hora_ini, hora_fim, dta_faturacao, cod_fatura)
Primary key nr_episodio
Candidate key cod_fatura

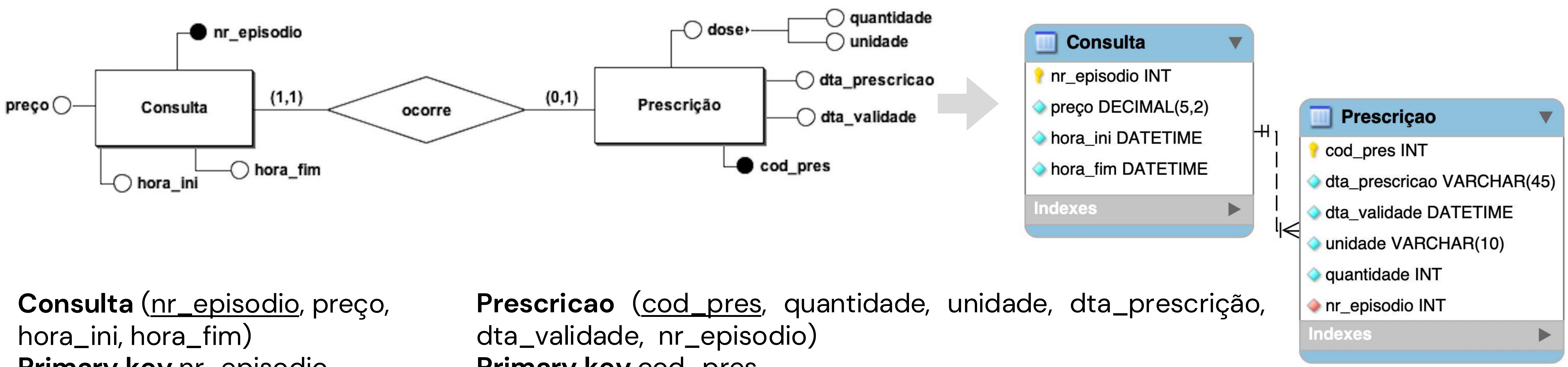
PHASE 4: Logical Modeling

Derive Relations

- **One-to-one (1:1) binary relationships**

(b) mandatory participation on one side of the 1:1 relationship;

- Copy of the parent entity's primary key placed in the table that represents the child entity.



Consulta (<u>nr_episodio</u>, preço, hora_ini, hora_fim)
Primary key nr_episodio

Prescricao (<u>cod_pres</u>, quantidade, unidade, dta_prescrição, dta_validade, nr_episodio)
Primary key cod_pres
Foreign key nr_episodio **references** Consulta(nr_episodio)

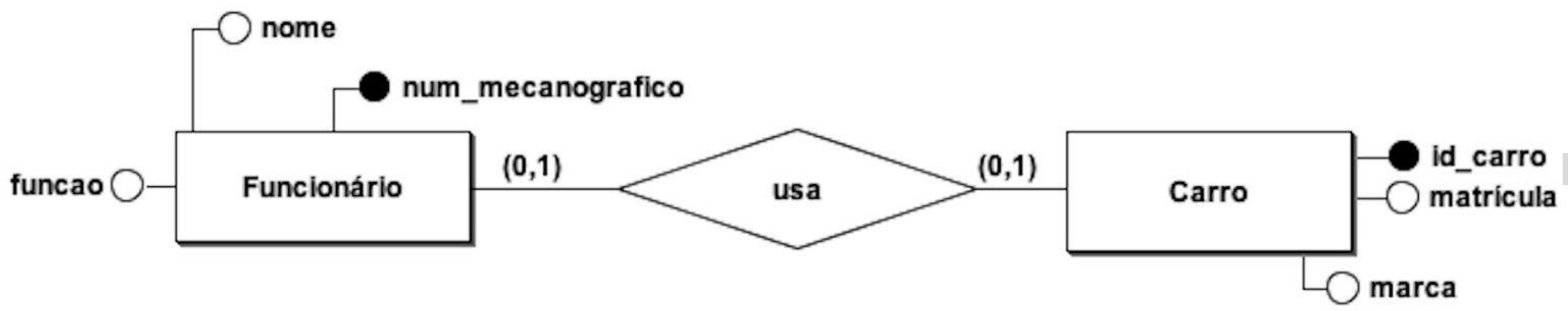
PHASE 4: Logical Modeling

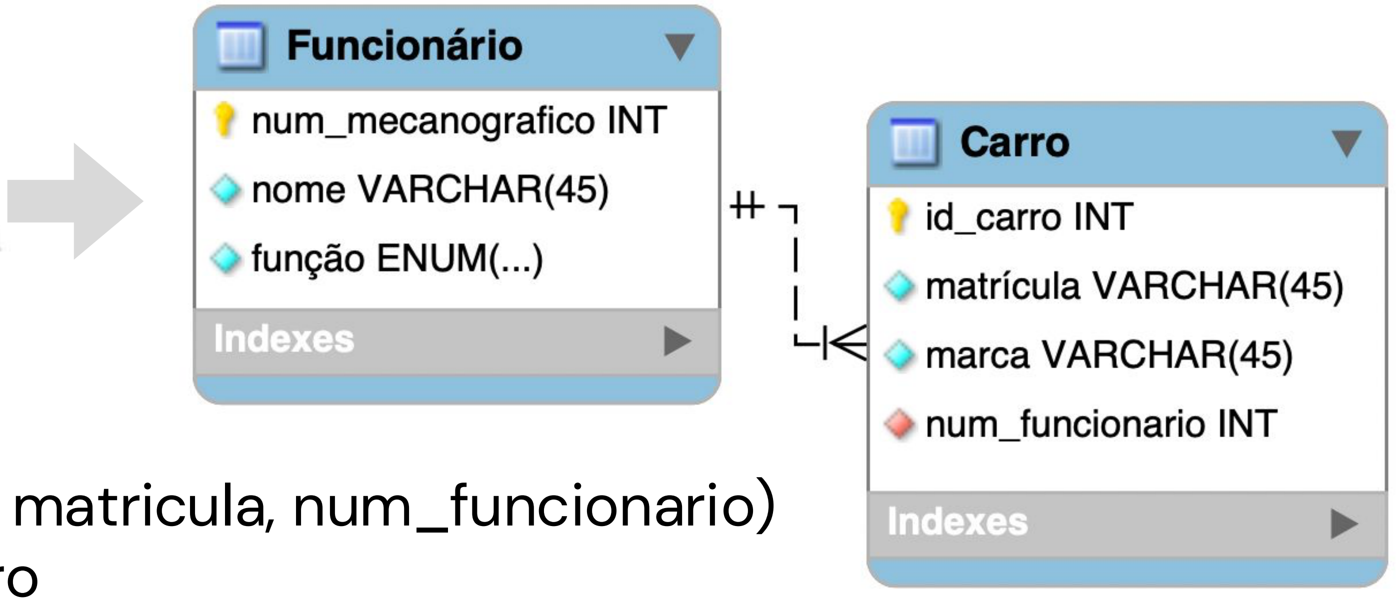
Derive Relations

- **One-to-one (1:1) binary relationships**

(c) optional participation on both sides of the 1:1 relationship.

Copy of the parent entity's primary key placed in the table that represents the child entity. The designation of parent and child entities is arbitrary unless more can be found out about the relationship.





Funcionário (num_mecanografico, nome, funcao)
Chave primária num_mecanografico

Carro (id_carro, marca, matricula, num_funcionario)
Chave primária id_carro
Chave estrangeira num_funcionario **referencia** Funcionário(num_mecanografico)

PHASE 4: Logical Modeling

Derive Relations

The recursive relationships of 1:N and N:M follow the rules of participation of a binary relationship of 1:N and N:M, respectively.

- **One-to-one (1:1) recursive binary relationships**

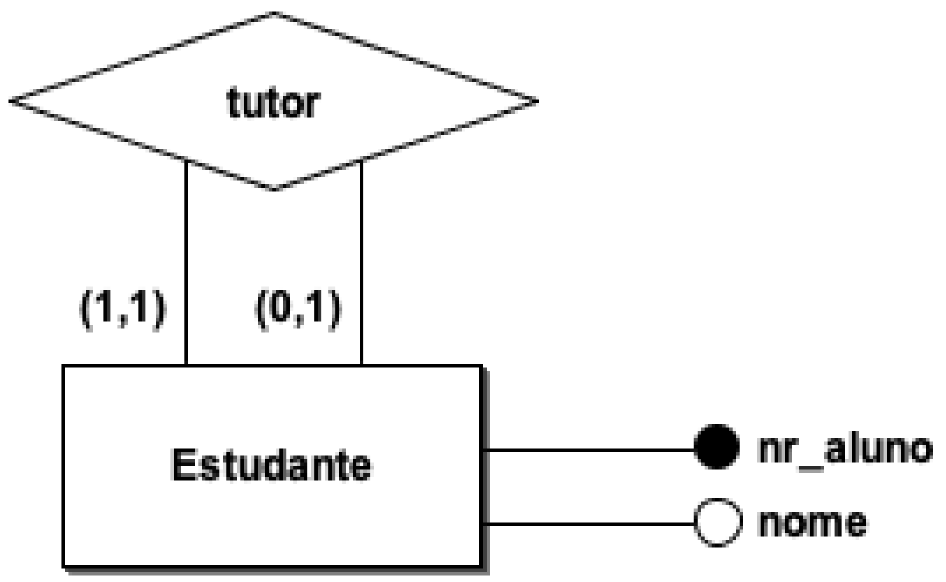
1:1 recursive relationships follow the rules:

- mandatory participation on both sides: unique table with a copy of the primary key to act as a foreign key that must be renamed for ease of interpretation and cannot be null (similar to the 1:N recursive relationship).
- optional participation of both sides: create a new table to represent the recursive relationship that would have only two attributes working with the primary key composed of the two primary keys that must be renamed for ease of interpretation and that also act as foreign keys (similar to the M:N recursive relationship).
- Mandatory participation on one side only: option to follow either of the two previous approaches.

PHASE 4: Logical Modeling

➔ Derive Relations

- **One-to-one (1:1) recursive binary relationships**



Estudante (nr_aluno, nome, tutor)
Primary key nr_aluno
Foreign key tutor **references** Estudante(nr_aluno)

ou

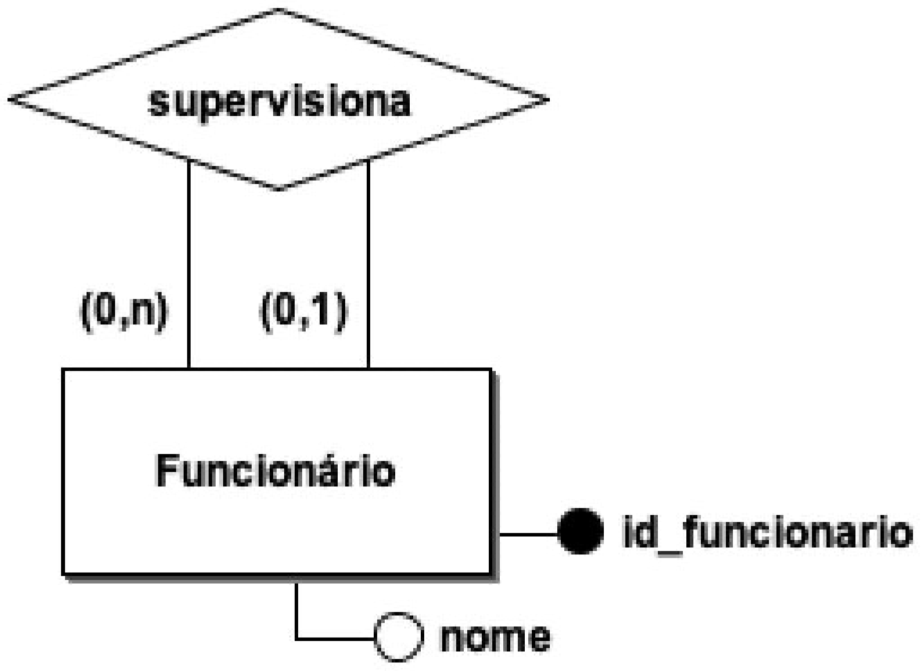
Estudante (nr_aluno, nome)
Chave primária nr_aluno

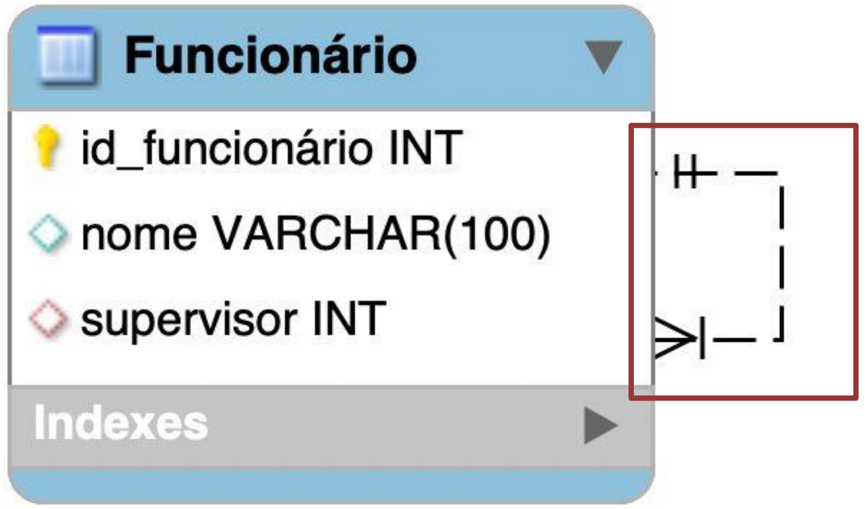
Tutor (nr_aluno, nr_aluno_tutor)
Primary key nr_aluno, nr_aluno_tutor
Foreign key nr_aluno_tutor **referencia** Estudante(nr_aluno)
Foreign key nr_aluno **references** Estudante(nr_aluno)

PHASE 4: Logical Modeling

Derive Relations

- **One-to-one (1:N) recursive binary relationships**





Funcionário (id_funcionario, nome, supervisor)
Primary key id_funcionario
Foreign key supervisor **references** Funcionário(id_funcionario)

PHASE 4: Logical Modeling

Derive Relations

- **Complex relationships**

- A complex relationship involves three or more entities. Although they are not very frequent in a conceptual scheme, this type of relationship arises in several practical situations in the development and use of databases.

- For each complex relationship, create a table to represent the relationship and include any attributes that are part of the relationship.

- We place a copy of the primary key(s) of the entities participating in the complex relationship in the new table, to act as foreign keys.

- The determination of the primary key depends on the cardinality of the complex relationship.

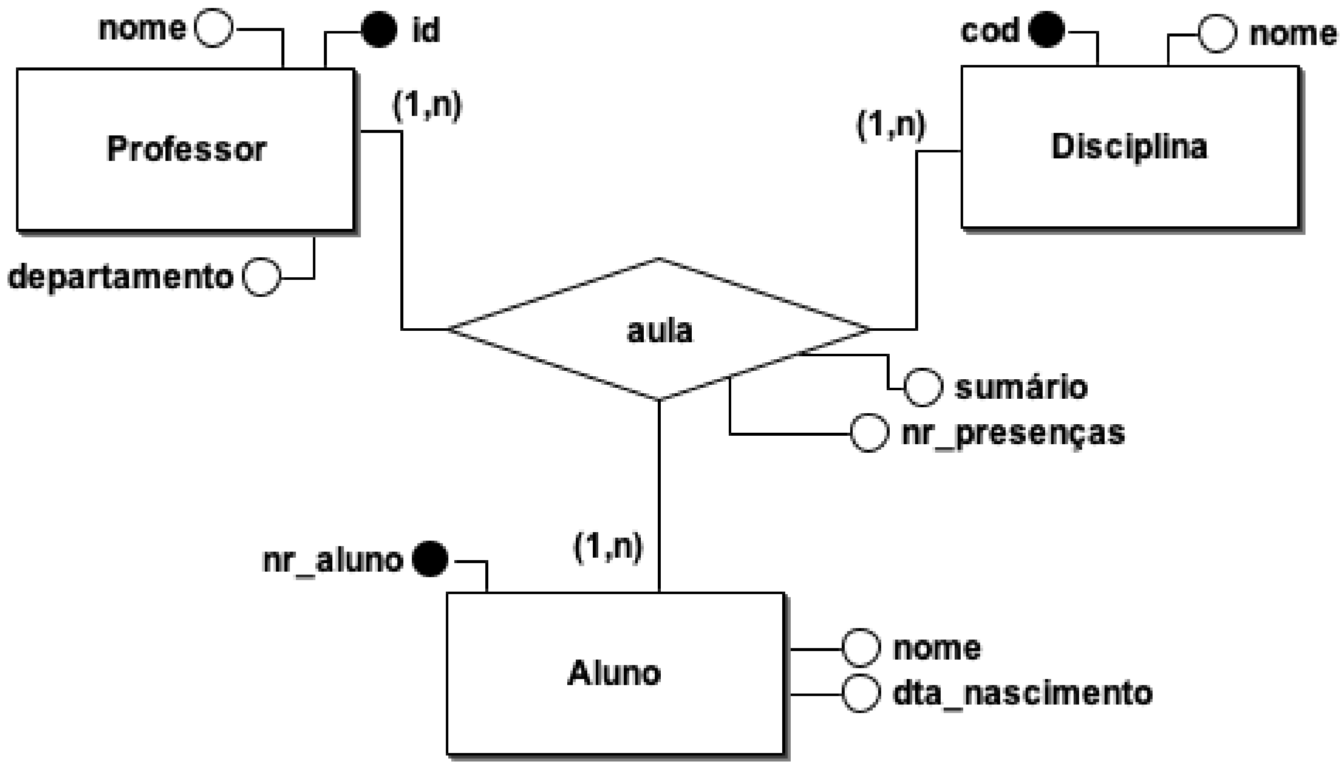
Cardinality	PK
N:M:P	Composed of the primary keys of the entities with cardinality **N**. *(idA, idB, idC)*
N:M:1	Composed of the primary keys of the entities with cardinality **N**. *(idA, idB)*
N:1:1	Composed of the primary key of the entity with cardinality **N**. *(idA)*
1:1:1	Composed of one of the pairs and the other two pairs must be **unique**.

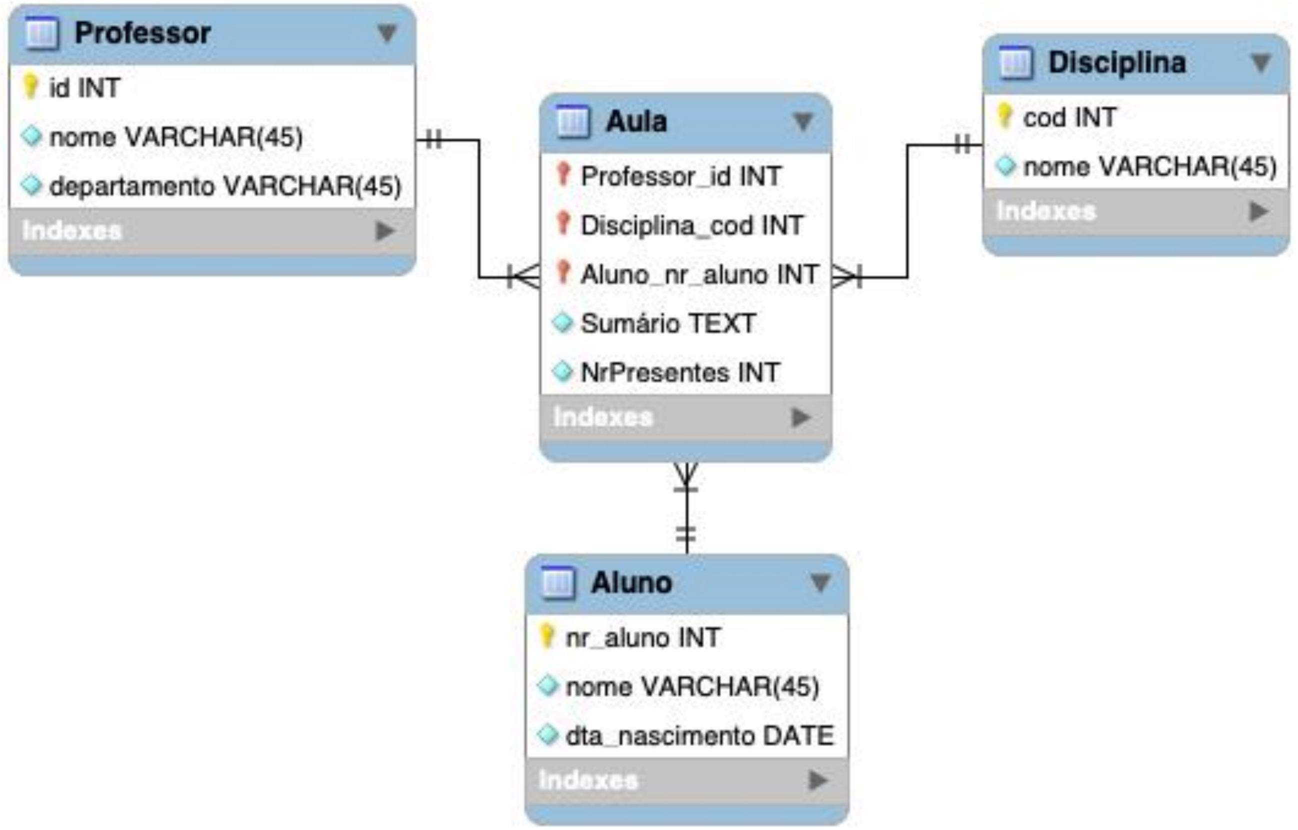
PHASE 4: Logical Modeling

➜ Derive Relations

- **Complex Relationships N:M:P**

– The new table has a primary key that is composed of the primary keys of the entities participating in the complex relationship.



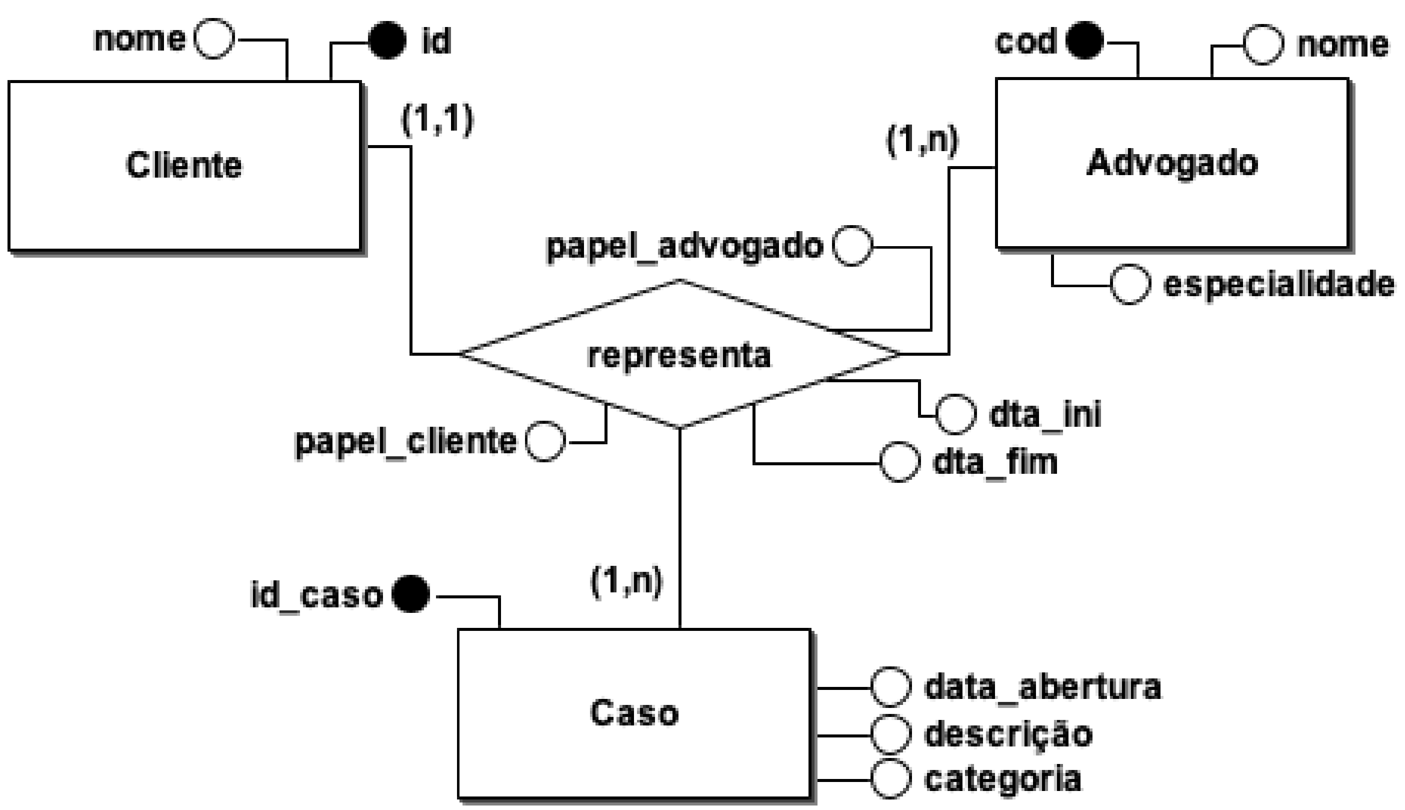


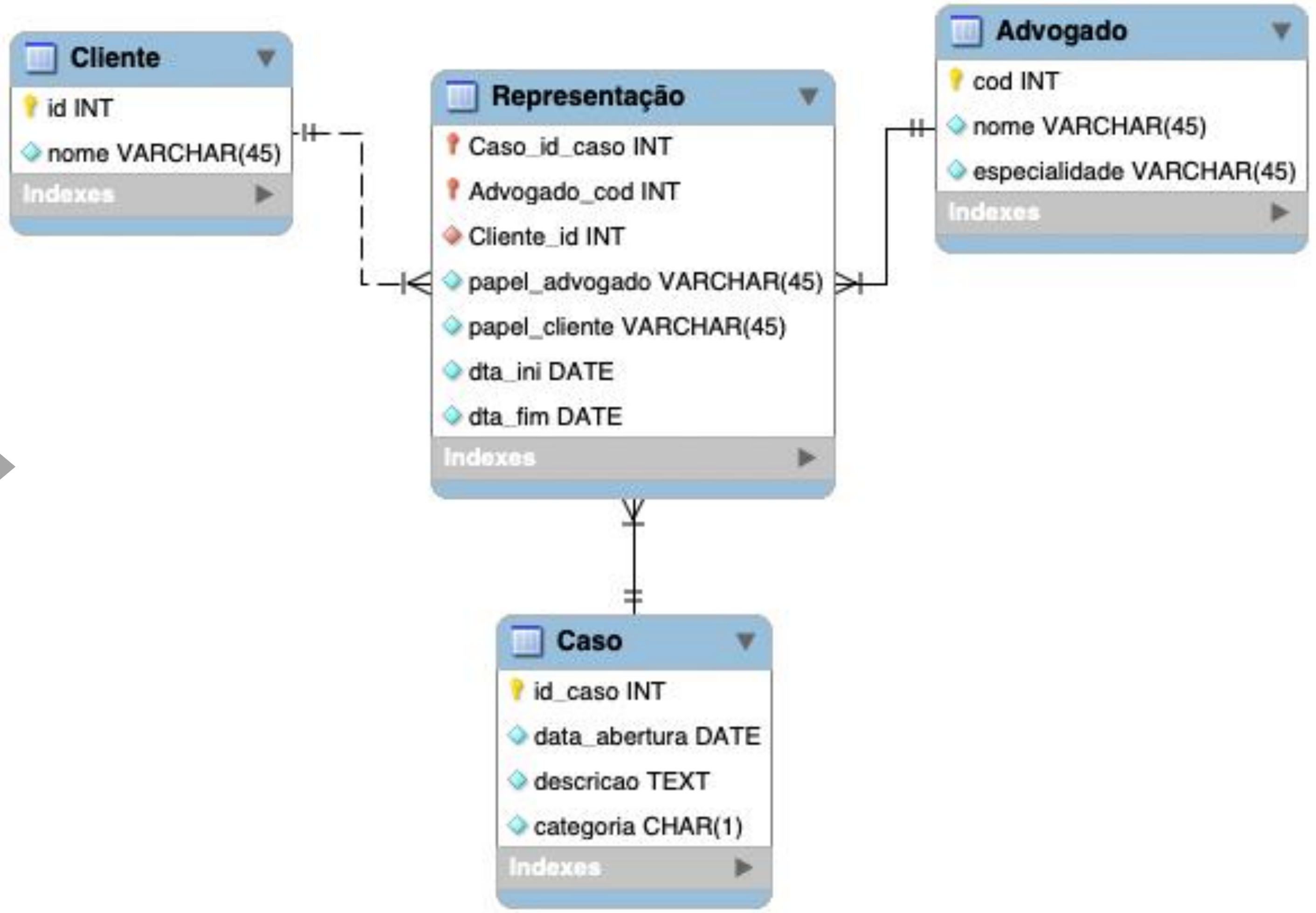
PHASE 4: Logical Modeling

Derive Relations

- **Complex relationships 1:N:M**

– AThe new table has a primary key composed of the primary keys of the entities that participate in the complex relationship with cardinality N.



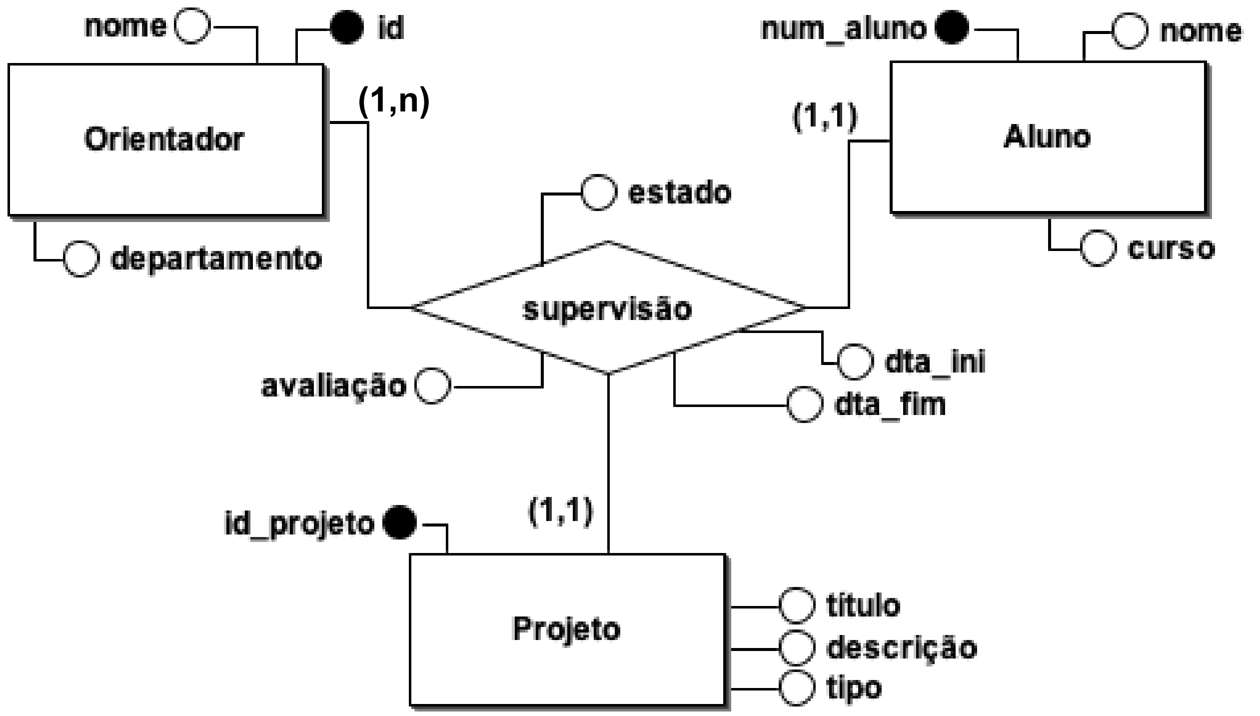


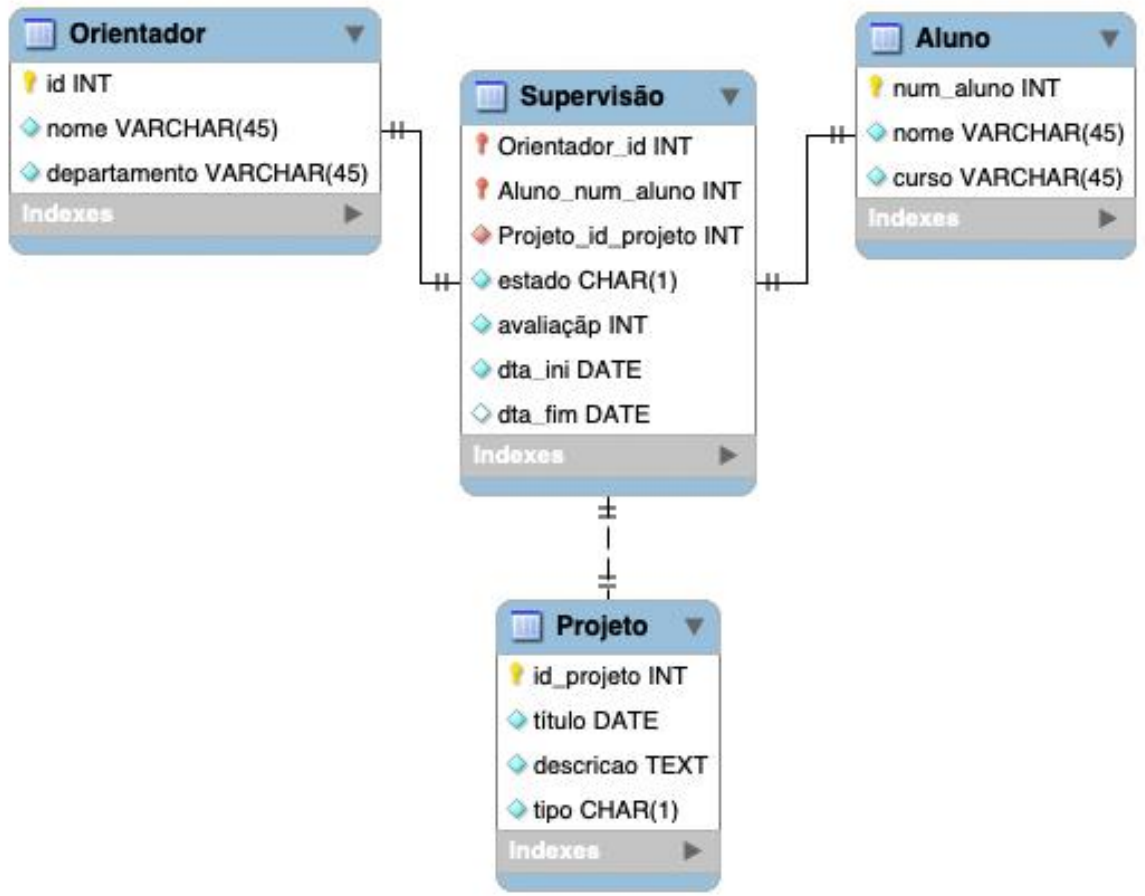
PHASE 4: Logical Modeling

Derive Relations

- **Complex 1:1:N relationships**

– The new table has a primary key composed of the primary key of the entity that participates in the complex relationship with cardinality N and the primary key of one of the other two entities, defined arbitrarily. In addition, the other pair must be unique.



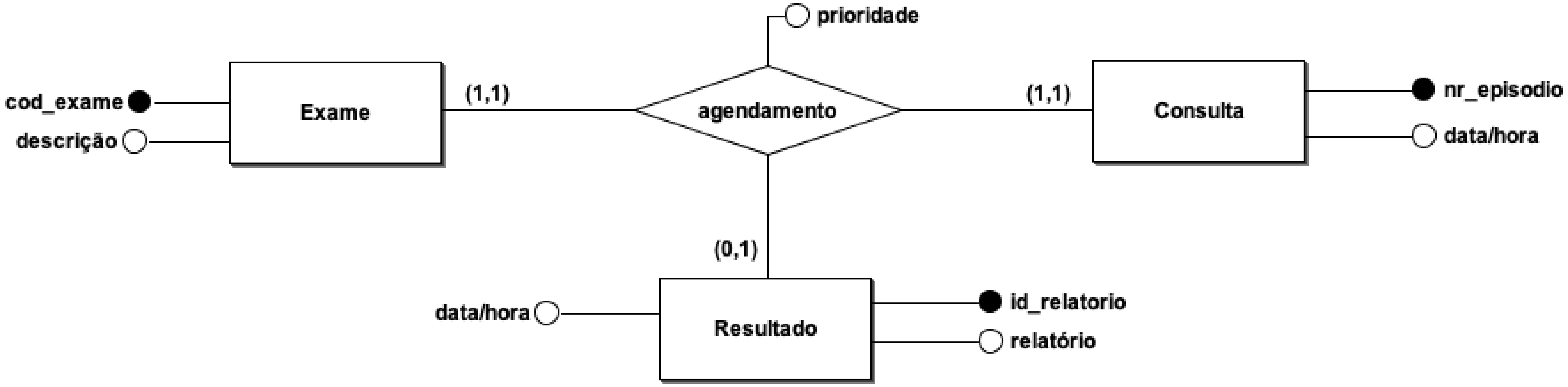


PHASE 4: Logical Modeling

Derive Relations

- **Complex 1:1:1 relationships**

– The new table has a primary key composed of the primary keys of two of the entities that participate in the complex relationship, which are arbitrarily defined. In addition, the other pair must be unique.



Agendamento (nr_episodio, id_relatorio, cod_exame, ...)
Primary key nr_episodio, id_relatorio
Foreign key nr_episodio **references** Consulta(nr_episodio)
Foreign key id_relatorio **references** Resultado(id_relatorio)
Foreign key cod_exame **references** Exame(cod_exame)

ou

Agendamento (nr_episodio, cod_exame, id_relatório,..)
Primary ke ynr_episodio, cod_exame

ou

Agendamento (id_relatorio, cod_exame, nr_episodio,...)
Primary key id_relatorio, cod_exame

PHASE 4: Logical Modeling

Derive Relations

- **Superclass/Subclass Relationships**

- Identify the superclass as the parent entity and the subclass as the child entity.

- The most appropriate representation of such a relationship depends on the number of:
  - disjunction restrictions and participation in the superclass/subclass relationship;
  - whether the subclasses are involved in distinct relationships;
  - number of participants in the superclass/subclass relationship.

PHASE 4: Logical Modeling

Derive Relations

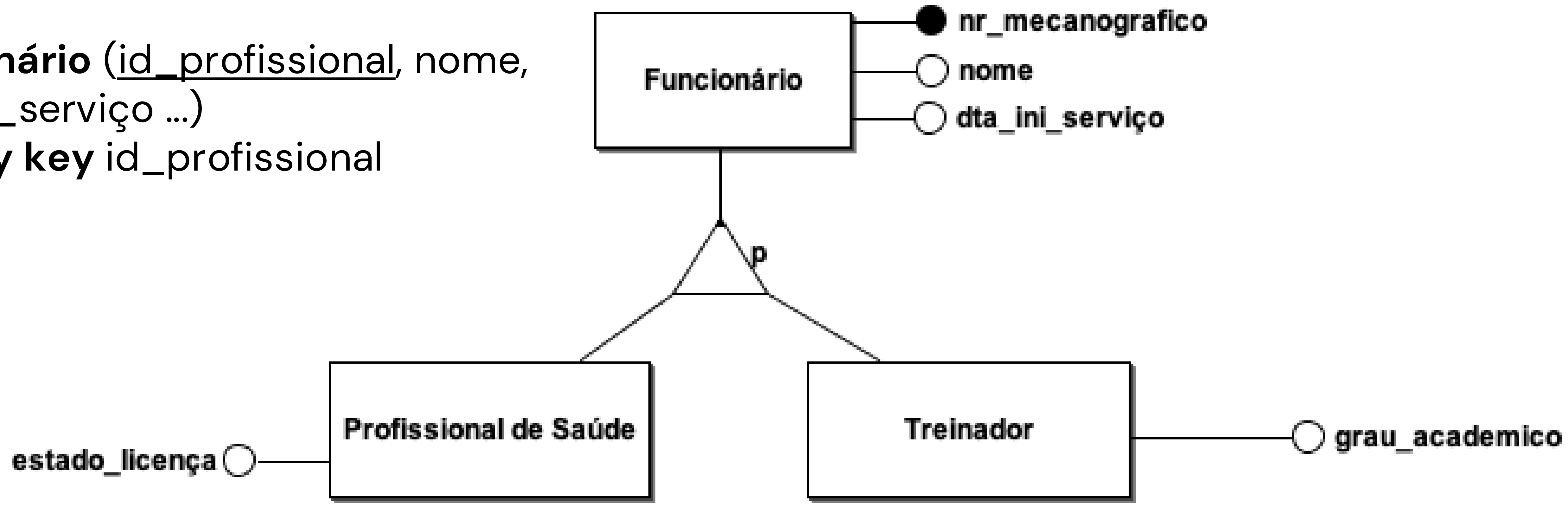
Participation Constraints	Disjointness Constraints	Required Relations
Mandatory	Non-disjoint {And}	**Single relation** with one attribute for each subclass *(flag)*
Mandatory	Non-disjoint {And}	**Two relations**: one relation for the **superclass** and one relation for **all subclasses** with one attribute for each subclass *(flag)*
Mandatory	Disjoint / Exclusive {Or}	**Many relations** *(one relation for each superclass/subclass combination)*
Mandatory	Disjoint / Exclusive {Or}	**Many relations** *(one relation for the superclass and one for each subclass)*

PHASE 4: Logical Modeling

➡ Derive Relations

Exemplo Optional/partial and exclusive

Funcionário (id_profissional, nome, dta_ini_serviço ...)
Primary key id_profissional



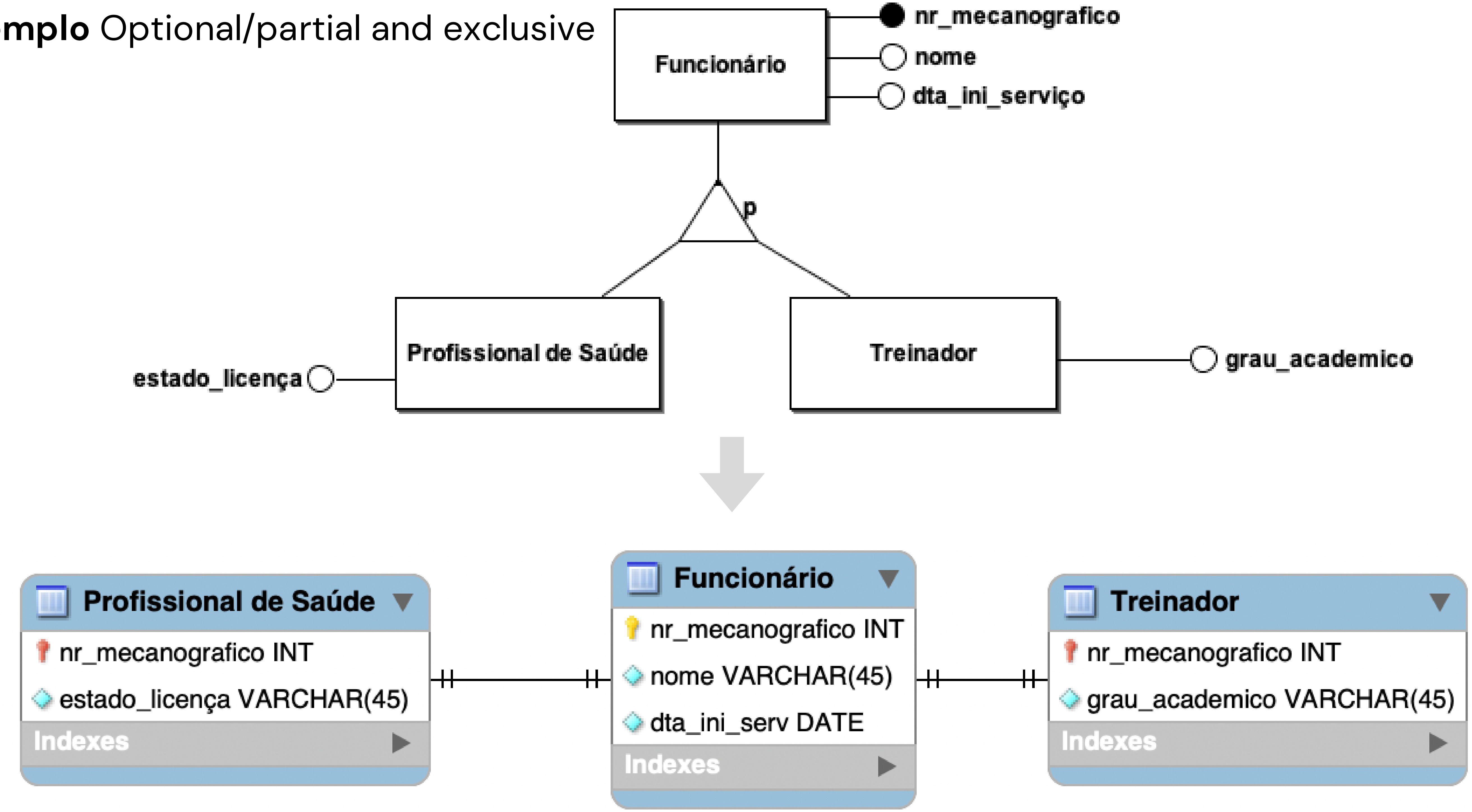
Profissional de Saúde (id_profissional, estado_licença)
Primary key id_profissional,
Foreign key id_profissional **references** Funcionário(id_profissional,)

Treinador (id_profissional, grau_academico)
Primary key id_profissional,
Foreign key id_profissional **references** Funcionário(id_profissional,)

PHASE 4: Logical Modeling

➔ Derive Relations

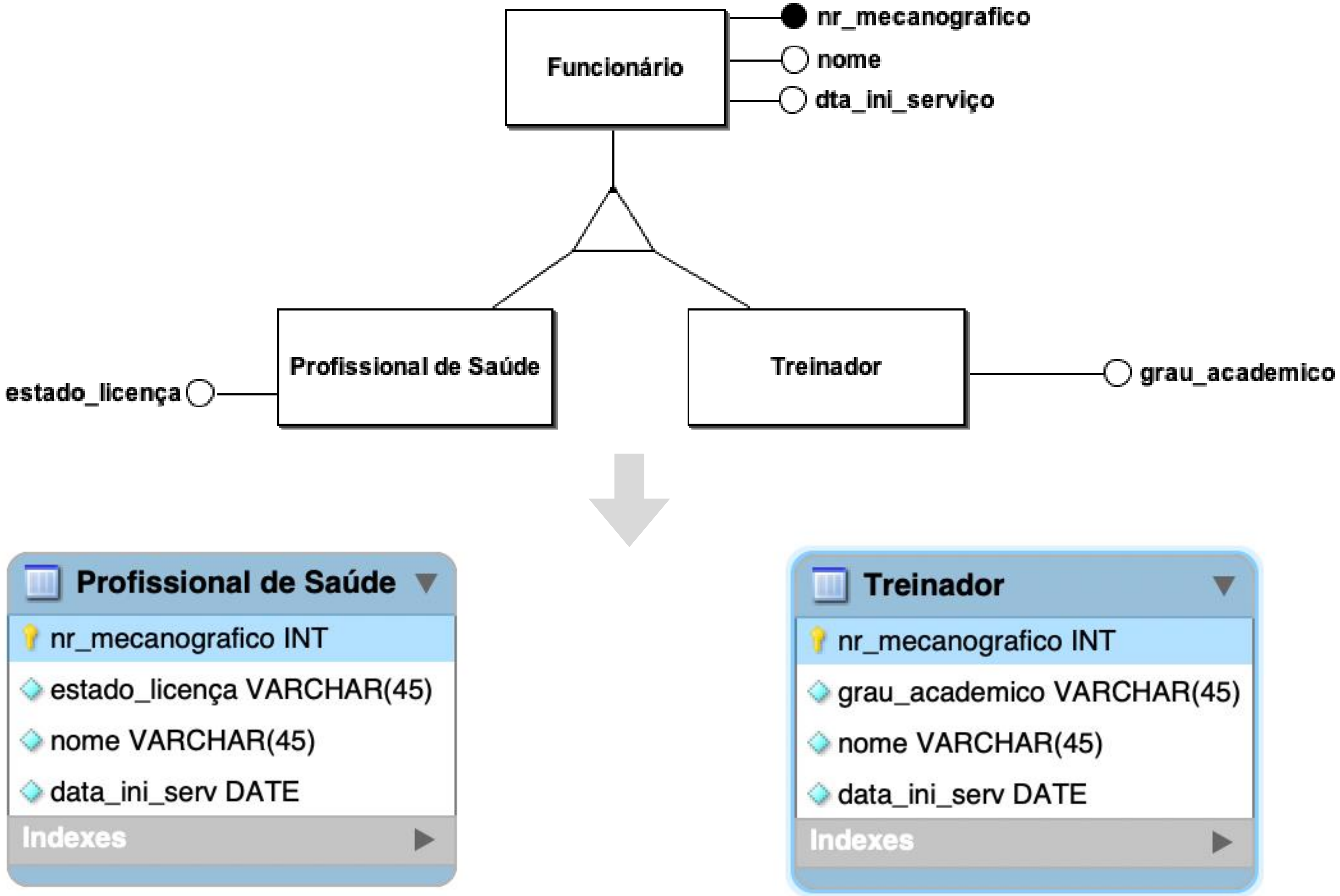
Exemplo Optional/partial and exclusive



PHASE 4: Logical Modeling

➜ Derive Relations

Exemplo Required/total and exclusive



PHASE 4: Logical Modeling

Integrity Constraints

- **Entity Integrity:** ensures that each record in a table is unique, usually through a **primary key**.
- **Referential Integrity:** ensures the consistency of relationships between tables by guaranteeing that **foreign keys correspond to valid primary keys**.
- **Domain Integrity:** ensures that the values inserted into attributes respect the defined **data types and constraints**.

PHASE 4: Logical Modeling

Integrity Constraints

Paciente				
nr_sequencial	nome	sexo	dta_nascimento	...
323431	Ana Luísa Dias Gomes	F	20/12/1990	
453347	José da Costa Silva	M	03/05/1975	
212423	Maria Leonor Ribeiro Barbosa	Fem	12/07/2000	
...	...	...	...	...

Referential Integrity ?

Domain Integrity

Consulta							
nr_episodio	id_pac	id_med	hora_ini	hora_fim	id_agenda	cod_proc	id_sec
12345678	212423	3456	2022-01-23 10:18:17	2022-01-23 10:38:27	123456789	P22	1212
14451643	453347	3224	2022-01-25 08:35:23	2022-01-25 09:00:12	223212434	P23	1598
14451643	212423	3371	2022-02-02 09:00:33	2022-02-02 09:15:20	345567811	NULL	1479
13415324	123456	3834	2022-02-04 12:34:11	2022-02-04 13:00:00	433212456	P22	1234
NULL	323431	NULL	2022-02-12 11:20:23	2022-02-12 11:52:33	387612392	P24	1176
...	...	...	...	...	...	...	...

Integrity of Entity

Integrity of Entity

Data requirement

Universidade do Minho

Ano Letivo: 2025/26

Databases

Logical Modeling

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